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TITLE: SketchSQL – AI DBMS APP

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ABSTRACT

1.1 Problem Background

Database schema design is a foundational step in every software development project, yet it remains time-consuming and error-prone for developers and students alike. Translating a conceptual data model into syntactically correct SQL DDL requires deep familiarity with database management systems, dialect differences between engines such as MySQL and PostgreSQL, and normalisation principles. Existing tools are either too simplistic, offering no AI-guided assistance, or too complex, demanding significant expertise before producing useful output. There is a clear and pressing need for an intelligent, accessible platform that bridges visual database modelling with automated code generation and contextual AI guidance, substantially lowering the barrier to effective schema design.

1.2 Objective of the Project

The primary objective of SketchSQL is to deliver a unified, local-first web platform that enables intuitive drag-and-drop entity-relationship diagram construction, real-time production-ready SQL DDL generation for multiple dialects, ORM code export for leading frameworks, and AI-powered schema assistance through natural language interaction — making professional-grade database design accessible to all skill levels.

1.3 Proposed Solution / System Overview

SketchSQL is a web-based visual database schema designer with a deeply integrated AI assistant. Users drag and drop tables onto a ReactFlow-powered canvas, define typed columns with constraints, and draw foreign-key relationships interactively. The application instantly generates production-ready SQL DDL for both MySQL and PostgreSQL dialects, alongside ORM models for Django, Prisma, and SQLAlchemy. An AI assistant powered by Azure OpenAI GPT-5.4 allows users to generate complete schemas from plain English descriptions, receive normalization analysis and targeted improvement suggestions, and engage in a multi-turn query assistant session. All data persists locally in the browser with no user accounts

required, and diagrams can be published as shareable read-only public links, making the tool equally suited for individual development and team review.

1.4 Key Features or Functionality

SketchSQL provides a visual ER diagram builder with customisable table nodes and typed relationship edges, real-time MySQL and PostgreSQL DDL generation with topological FK ordering, ORM code generation for Django Models, Prisma Schema, and SQLAlchemy, SQL import for reverse-engineering existing schemas, AI-powered schema generation from natural language, normalisation analysis with actionable recommendations, a multi-turn query-writing assistant with full schema context, auto-save and named diagram saves via localStorage and IndexedDB, PNG, PDF, and JSON export, shareable public read-only diagram links, undo/redo with a 50-state history stack, snap-to-grid layout, and dark/light theme support.

1.5 Technology / Method Used

The frontend is built with React 18, ReactFlow 11, Zustand for global state management, Monaco Editor for syntax-highlighted SQL output, and Tailwind CSS for theming. The backend uses Python 3 with FastAPI and Uvicorn. AI capabilities are delivered through Azure OpenAI (GPT-5.4) via the AzureOpenAI SDK client. Diagram data persists locally using localStorage and IndexedDB, with MongoDB used exclusively for public-share snapshots.

1.6 Expected Outcome & Benefits

SketchSQL is expected to significantly reduce the time required to design, document, and prototype database schemas. Students and developers benefit from instant visual feedback, accurate multi-dialect SQL generation, and AI-guided design assistance without requiring deep prior SQL expertise. The platform promotes database normalisation best practices, reduces manual coding errors, and accelerates the handoff between schema design and application development by providing ready-to-use ORM models alongside standard DDL — ultimately improving productivity and code quality across academic and professional projects.

1.7 Original Contribution Statement

This work presents an original integration of visual ER diagram construction, multi-dialect SQL generation, ORM code synthesis, and conversational AI schema assistance within a single, local-first web application.